

Clustering-Enhanced Next POI Recommendation Using Hierarchical Graph Neural Networks

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Abstract

Next Point of Interest (POI) recommendation is a crucial task in location-based services, aimed at predicting the next location a user is likely to visit. This task is inherently connected to the problem of personalized route planning, where the objective is to determine an optimal route that aligns with the user's preferences and past behavior. By accurately predicting the next POI, we can effectively build personalized routes that cater to the individual needs of users. In this paper, we propose an advanced model that leverages two Graph Neural Networks (GNNs) to handle spatial and temporal data separately. Furthermore, we enhance this architecture by incorporating user clustering, providing an additional layer of user information through the embedding of cluster centroids and the most relevant users in it. This approach allows us to capture shared POI preferences among users within the same cluster, improving the accuracy of the next POI prediction and consequently enhancing the quality of personalized routes. Experiments on real-world datasets demonstrate that our model gets better results than state-of-the-art methods in various evaluation metrics, showcasing its effectiveness in understanding and predicting user mobility patterns.

Keywords

Next Point of Interest, Graph Neural Network, Personalized Route, Deep Learning

1. Highlights

- HMT-GRN-C and HMT-GRN-C-TK novel architectures to work on Next Point of Interest Recommendation.
- Spatial and Temporal Graph Neural Networks.
- Hierarchical Beam Search on POIs.
- Clusters on users.
- Higher accuracy on predictions.

2. Introduction

In recent years, the large volume of check-in records provided by location-based services, such as Foursquare, has greatly promoted research on next Point-of-Interest (POI) recommendation [1, 2, 3]. These methods aim at modeling sequential check-in behaviors and predicting time-sensitive POI visits, emphasizing that learning user preferences from past and current check-in behaviors is fundamental for boosting the next POI prediction[1, 4, 5].

Next POI recommendation involves predicting the next location a user will visit based on their historical movement data and contextual information. This task is not only important for enhancing user experience in location-based services but also plays a crucial role in personalized route planning. By understanding and predicting a user's next destination, we can create more personalized and efficient travel routes that cater to the user's preferences and needs, thereby improving overall satisfaction and engagement[6, 7].

Many next POI recommenders focus on modeling users' current check-ins (i.e., short-term behaviors) under the assumption that users' next movements are highly correlated with their latest check-in behaviors. For instance, the pioneering work FPMC-LR models two successive check-ins with a localized region constraint incorporated[1]. Subsequent studies derived from metric embedding and Word2Vec have aimed to model sequential check-ins and learn transition regularities by capturing the local co-occurrence of

POIs[1, 4, 8]. More recently, recurrent neural networks (RNN) and their variants, such as ST-RNN and ARNN, have gained attention for capturing users' current preferences by modeling entire sequential information rather than pairs of POIs within a trajectory[1, 9, 3].

The task of next POI recommendation is closely related to personalized route planning, where the goal is to determine a route that best fits an individual user's preferences and behavior. By accurately predicting the next POI, it becomes feasible to construct personalized routes that not only cater to the immediate next location but also optimize the overall travel experience based on user preferences and historical data. This makes next POI recommendation a critical component in the broader scope of personalized route planning.

Several methodologies have been proposed to tackle the next POI recommendation task, ranging from traditional collaborative filtering approaches to advanced deep learning techniques. Collaborative filtering methods, such as matrix factorization[10], aim to capture user-item interactions based on historical data. However, these methods often struggle with sparsity issues and fail to consider the sequential nature of user movements.

To address these limitations, more recent approaches have leveraged sequential models like Recurrent Neural Networks (RNNs) and their variants (e.g., LSTMs and GRUs) to model temporal dependencies in user check-in data[3, 6]. These models can capture the sequential patterns of user movements, but they may still fall short in capturing complex spatio-temporal dependencies.

Graph-based methods have also gained traction in this domain, as they can naturally model the relationships between POIs and users in a graph structure. Graph Neural Networks (GNNs) have been used to capture both the spatial and temporal dynamics of user movements[11, 12]. Additionally, incorporating contextual information, such as time and weather, can further enhance the prediction accuracy[13].

In this paper, we propose an advanced model that leverages two Graph Neural Networks (GNNs) to handle spatial and temporal data separately. Furthermore, we enhance this architecture by incorporating user clustering, providing an additional layer of user information through the embedding of cluster centroids. This approach captures shared POI preferences among users within the same cluster, improving

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the accuracy of the next POI prediction. Extensive experiments on real-world datasets demonstrate that our model get slightly better results than state-of-the-art methods in various evaluation metrics, showcasing its effectiveness in understanding and predicting user mobility patterns.

3. Related Work

The Next Point of Interest (POI) recommendation task is a critical component of location-based services, aimed at predicting the next location a user is likely to visit based on their historical check-in data and contextual information. This task has garnered significant attention due to its practical applications in enhancing user experience, optimizing travel routes, and improving the relevance of location-based recommendations.

3.1. The Task of Next POI Recommendation

The primary objective of the Next POI recommendation task is to forecast a user's subsequent check-in location by analyzing their historical movement patterns and contextual data such as time, day of the week, and current location. This task is inherently challenging due to the complex and dynamic nature of human mobility patterns, which are influenced by various factors including personal preferences, social interactions, and environmental contexts.

Formally, the Next POI recommendation problem can be defined as follows: Given a user's check-in sequence $S_u = [l_1, l_2, \dots, l_t]$ up to time t , where l_i represents the location visited at time i , the goal is to predict the next location l_{t+1} . The recommendation system must consider spatial and temporal dependencies in the user's historical data, as well as any available contextual information.

To tackle this problem, various approaches have been proposed in the literature, ranging from traditional collaborative filtering techniques to advanced deep learning models. The effectiveness of a Next POI recommendation system is typically evaluated based on metrics such as accuracy, precision, recall, and mean reciprocal rank (MRR), which measure the system's ability to correctly predict the user's next location within a ranked list of candidate locations.

3.2. Methods in Literature

Research on Next POI recommendation has explored a variety of methods to model and predict user check-in behaviors. One line of research assumes that users' next check-in behaviors are highly influenced by their recent visits. Markov-based methods have been introduced to learn the transition probability between successive check-ins[4, 14].

Matrix Factorization: Collaborative filtering techniques like matrix factorization have been widely used to model user-item interactions. These methods decompose the user-item interaction matrix into lower-dimensional representations, capturing latent factors that explain user preferences[10]. While effective in many cases, matrix factorization methods often struggle with sparsity issues and do not account for the sequential nature of check-ins.

Sequential Models: To address the temporal aspect of user check-ins, sequential models such as Recurrent

Neural Networks (RNNs) and their variants (e.g., Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs)) have been employed. These models can capture the dependencies between successive check-ins by learning from the sequence of past behaviors[3, 6]. For example, ST-RNN (Spatial-Temporal Recurrent Neural Networks) models the spatiotemporal regularities in user movements, and SERM (Spatial-Temporal Recurrent Model) extends this approach by incorporating additional contextual information[15, 16].

Geographical Influence: One important aspect of Next POI recommendation is the geographical influence on user mobility patterns. Models such as GeoMF (Geographical Matrix Factorization) [5] incorporate geographical proximity and spatial constraints into the recommendation process. This approach assumes that users are more likely to visit nearby POIs and integrates this information into a matrix factorization framework to enhance prediction accuracy.

Temporal Dynamics: The temporal dynamics of user check-in behavior play a crucial role in predicting the next POI. Time-LSTM [17] is an extension of the Long Short-Term Memory (LSTM) network that incorporates time intervals between check-ins to model the temporal dynamics more effectively. By explicitly considering the time gaps, Time-LSTM can better capture the temporal dependencies and improve the recommendation performance.

Context-Aware Models: Contextual information such as weather conditions, social interactions, and user preferences can significantly enhance the quality of Next POI recommendations. Context-Aware POI Recommendation (CAPR) [18] leverages contextual factors by integrating them into a unified framework along with user mobility data. This approach enables the model to make more personalized and contextually relevant recommendations.

Attention Mechanisms: Attention mechanisms have been introduced to improve the performance of sequential models by focusing on the most relevant parts of the input sequence. The Attentive Recurrent Network (ARN) [3] applies an attention mechanism on top of an RNN to dynamically weigh the importance of past check-ins. This helps the model to prioritize important check-ins and enhances the accuracy of Next POI recommendations.

Adversarial Learning: Adversarial learning techniques have been explored to enhance the robustness and generalization of Next POI recommendation models. The Adversarial Spatial-Temporal Network (ASTN) [19] employs a generative adversarial network (GAN) framework to generate plausible POI sequences that mimic real user behaviors. By training the recommendation model with these adversarial examples, ASTN improves its ability to handle noisy and incomplete check-in data.

Graph Neural Networks: Graph Neural Networks (GNNs) have gained popularity for their ability to model relational data. The Dynamic POI Graph Convolutional Network (DPGCN) [20] constructs a dynamic graph of POIs based on user check-in sequences and applies graph convolutions to capture the spatial dependencies. This approach leverages the graph structure to improve the representation of POIs and enhances the recommendation accuracy.

4. Method Proposed

Our proposed model builds on these Graph-based Methods by leveraging two Graph Neural Networks (GNNs) to handle spatial and temporal data separately and incorporating user clustering to enhance prediction accuracy. This approach captures shared POI preferences among users within the same cluster, providing more accurate next POI predictions. In this section we propose three architectures: a Baseline (HMT-GRN) and two novelties (HMT-GRN-C, HMT-GRN-C-TK). Both of them rely on the usage of Recurrent Neural Networks (RNNs) and Graph Neural Networks (GNNs).

4.1. HMT-GRN

The HMT-GRN (Hierarchical Multi-Task Graph Recurrent Network) [4] architecture is designed to handle the complexity of Next POI recommendation by leveraging both spatial and temporal data. The architecture consists of several key components that work together to capture dynamic user behaviors and evolving preferences over time.

The core of the HMT-GRN is a graph-based recurrent neural network (RNN) that processes sequential user check-in data. This network captures both sequential dependencies and global POI-POI relationships by utilizing a hierarchical structure divided into two main layers: a region-level task and a POI-level task.

4.1.1. Region-Level Task

The region-level task aims to capture broader geographical preferences by clustering similar POIs into regions. This step helps reduce computational complexity and enhances the interpretability of the model. The clustering of POIs is based on their spatial proximity, typically using methods such as geohashing.

4.1.2. POI-Level Task

The POI-level task focuses on the finer details within each region, modeling user-POI interactions to predict the next POI within the identified region. This involves constructing a POI-POI graph for each region, where nodes represent POIs and edges represent transitions between them based on historical check-in data.

4.1.3. Graph Recurrent Network (GRN)

The Graph Recurrent Network (GRN) integrates both the region-level and POI-level graphs to make predictions. The GRN module extends the Dimensional GAT (DGAT) variant by incorporating a recurrent structure that allows the model to learn temporal sequences of check-ins. The hidden state h_t at time t is updated based on the current input x_t and the previous hidden state h_{t-1} :

$$h_t = \text{LSTM}(x_t, h_{t-1})$$

For the spatial component, the GRN employs graph attention layers to aggregate information from neighboring nodes. The hidden state $h_v^{(l+1)}$ of node v at layer $l+1$ is computed as:

$$h_v^{(l+1)} = \sigma \left(\sum_{u \in \mathcal{N}(v)} \alpha_{vu}^{(l)} W^{(l)} h_u^{(l)} \right)$$

where $h_v^{(l)}$ is the hidden state of node v at layer l , $\mathcal{N}(v)$ is the set of neighbors of node v , $\alpha_{vu}^{(l)}$ is the attention coefficient for the edge between nodes v and u at layer l , and $W^{(l)}$ is the layer-specific trainable weight matrix. The attention coefficients $\alpha_{vu}^{(l)}$ are computed as:

$$\alpha_{vu}^{(l)} = \frac{\exp \left(\text{LeakyReLU} \left(a^{(l)\top} [W^{(l)} h_v^{(l)} \| W^{(l)} h_u^{(l)}] \right) \right)}{\sum_{k \in \mathcal{N}(v)} \exp \left(\text{LeakyReLU} \left(a^{(l)\top} [W^{(l)} h_v^{(l)} \| W^{(l)} h_k^{(l)}] \right) \right)}$$

where $a^{(l)}$ is a learnable weight vector and $\|$ denotes concatenation. This attention mechanism allows the GRN to focus on the most relevant neighbors when updating the node representations.

This combination of recurrent and graph attention mechanisms enables the GRN to effectively capture both temporal and spatial dependencies in the check-in sequences.

4.1.4. Hierarchical Beam Search (HBS)

To efficiently generate the first-K next POI predictions, the model employs Hierarchical Beam Search (HBS). HBS operates by first selecting the most promising regions based on the region-level graph and then narrowing down to the specific POIs within those regions using the POI-level graph. This two-stage process helps reduce the search space and improves prediction accuracy.

$$P(G_r, G_p | h_t, l_t) = P(G_r | h_t) \cdot P(G_p | G_r, h_t, l_t)$$

where $P(G_r | h_t)$ is the probability of selecting a region given the hidden state, and $P(G_p | G_r, h_t, l_t)$ is the probability of selecting a POI within the chosen region.

4.2. HMT-GRN-Clustering

We introduce a novel embedding technique for each user. We construct an embedding vector for each user, where the dimensionality of the vector corresponds to the total number of POIs, denoted as n . Each element in the vector represents a specific POI. The value of an element is set to m if the user has visited the POI m times, and is set to 0 if the user has never visited that POI.

Formally, let $\mathbf{v}_u$ be the embedding vector for user u :

$$\mathbf{v}_u = [v_1, v_2, \dots, v_n]$$

where

$$v_i = \begin{cases} m & \text{if user } u \text{ has visited POI } i \text{ } m \text{ times} \\ 0 & \text{if user } u \text{ has never visited POI } i \end{cases}$$

After constructing the embedding vectors, we apply different clustering algorithms to group users with similar POI visitation patterns. The objective of the clustering algorithm is to partition the users into k clusters $C_1, C_2, \dots, C_k$ such that the within-cluster sum of squares (WCSS) is minimized. The clustering process is defined by:

$$\min \sum_{j=1}^k \sum_{\mathbf{v}_u \in C_j} \|\mathbf{v}_u - \mu_j\|^2$$

where μ_j is the centroid of cluster C_j .

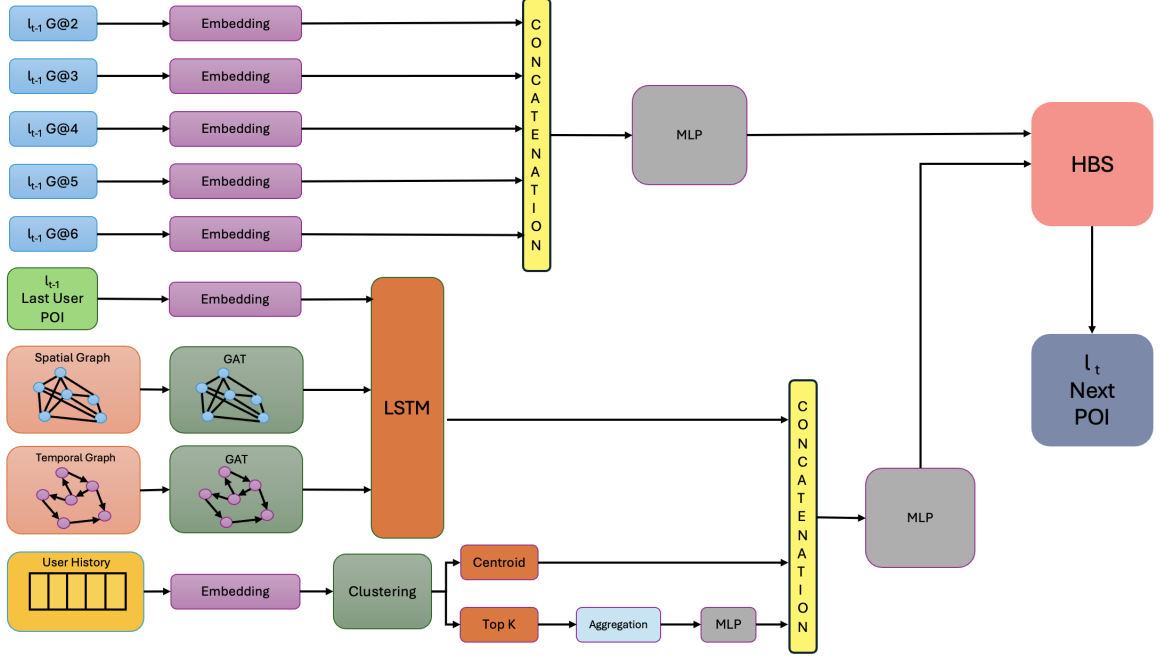


Figure 1: Illustration of the HMT-GRN-C-Top-K model that includes shared feature learning by our GRN module, exploitation of the User history using clustering and followed by the multi-task learning of next regions and POI, then performing our HBS and selectivity layer.

The embedding of a centroid $\mu_j \in \mathbb{R}^n$ is representative of the users that belong to it.

This provides the model with additional information about the group behavior of users with similar POI visit patterns.

Let $\mathbf{c}_u$ be the centroid embedding for the cluster to which user u belongs. The final input embedding $\mathbf{e}_u$ for user u is given by:

$$\mathbf{e}_u = [\mathbf{u}, \mathbf{c}_u]$$

where $\mathbf{e}_u$ is a concatenation of the user's individual embedding $\mathbf{u}$ and the centroid embedding $\mathbf{c}_u$.

By incorporating the centroid embeddings, our model gains additional insights into the collective behavior of users within the same cluster. This can enhance the prediction accuracy by leveraging the shared patterns among users in the same cluster.

For clustering, we utilized both the k-means and Gaussian Mixture model algorithms.

4.3. HMT-GRN-Clustering Top-k

To further enhance the information passed to the model regarding user behavior based on similar users, we have introduced an additional embedding.

Given a user u and his cluster C_i , we not only include the embedding of the centroid $\mathbf{c}_u$ but also select the embeddings of the K most similar users within the same cluster.

By selecting $v_k \in V_K$ embeddings of the M most similar users, we increase the likelihood of selecting a next POI for user u that is more likely to be their next choice.

To select the embeddings of the users most similar to the selected user, we take all embedding vectors $v_k \in \mu_j$ and compare these vectors with the embedding of user u , i.e., v_u . We calculate the cosine similarity between these vectors as follows:

$$\text{cosine_similarity}(\mathbf{u}_i, \mathbf{u}_j) = \frac{\mathbf{u}_i \cdot \mathbf{u}_j}{\|\mathbf{u}_i\| \|\mathbf{u}_j\|}$$

Once all cosine similarities are calculated, we sort them from highest to lowest and select the top K embeddings, thereby obtaining the embeddings of the users most similar to the given user.

4.3.1. Aggregation

Once the top K user embeddings are obtained, we use two alternative methods to aggregate the information from these embeddings.

Concatenation: We concatenate the embeddings of the top- k users and then feed the concatenation to an MLP.

Average: We compute the average of the embeddings of the top- k users and then feed the average to an MLP.

4.3.2. Formalization of the Problem with Clustering

Given a user u , his embedding v_u , and his cluster C_i , we define the following steps to select the top- k similar users within the cluster:

1. Calculate the cosine similarity between the embedding of user u and the embeddings of all other users in the cluster C_i :

$$\text{cosine_similarity}(v_u, v_k) = \frac{v_u \cdot v_k}{\|v_u\| \|v_k\|}$$

2. Rank the users in the cluster based on the cosine similarity scores in descending order.

3. Select the top- k users with the highest cosine similarity scores.

4. Aggregate the embeddings of these top- k users using either concatenation or averaging and feed them to a MLP:

- **Concatenation:**

$$\text{MLP}(\text{cat}(v_{u1}, v_{u2}, \dots, v_{uk}))$$

- **Average:**

$$\text{MLP}(\text{avg}(v_{u1}, v_{u2}, \dots, v_{uk})) = \text{MLP}\left(\frac{1}{k} \sum_{i=1}^k v_{ui}\right)$$

4.4. Functioning of the Model

In our extended model, the objective is to refine the predictions by incorporating additional embeddings that represent user similarity. Here is a formalized explanation of the model architecture and functioning:

Overview

The model processes the input data through several key components, each contributing to the final prediction of the next Point of Interest (POI). The main components are:

- **User Embeddings:** Represents user preferences and behaviors.
- **ST-GAT (Spatial-Temporal GAT):** Captures relationships between different POIs through spatial and temporal graphs.
- **LSTM (Long Short-Term Memory):** Tracks the sequence of POIs visited by the user over the week.
- **Clustering Embeddings:** Embeddings of the clusters to which users belong, including embeddings of the top- k most similar users.

Detailed Processing

1. Embedding Layer:

- Each POI is embedded using a neural network embedding layer.

2. Graph Attention Networks (GATs):

- Spatial GAT and Temporal GAT process the spatial and temporal relations of POIs, respectively.
- For each POI in the time sequence, the model identifies its neighbors in the spatial and temporal graphs and computes the attention-weighted embeddings.

3. LSTM Layer:

- The LSTM layer integrates the sequential information of POIs visited by the user during the week, combining them with the output of the ST-GAT.

4. Cluster and Top-k User Embeddings:

- User and cluster embeddings are concatenated with the LSTM output.
- Top- k user embeddings are averaged/concatenated and added to the final embedding.

5. Prediction Layers:

- The final embeddings are passed through multiple prediction layers, including MLPs, to predict the next POI and the geohash of the next location.

4.5. Loss Functions

The total loss $\mathcal{L}$ of the baseline model is computed as follows:

For each task $t_k \in \{G@2, G@3, \dots, G@6, POI\}$, where $\{G@2, G@3, \dots, G@6\}$ are the region-level tasks and $\{POI\}$ is the POI level task, the cross-entropy loss is defined as $\mathcal{L}_{t_k} = -\sum_{v=1}^{N_{\text{train}}} \log(p_{b_v})$, where p_{b_v} is the predicted probability of the ground truth next POI or $G@P$, depending on the task t_k , for the v -th training sample, and N_{train} is the total number of training samples. The overall supervised loss $\mathcal{L}$ is computed as the average of the losses for each task, ensuring that no single task biases the learning process:

$$\mathcal{L} = \frac{1}{|TK|} \sum_{t_k} \mathcal{L}_{t_k}$$

In our variants, the loss function is slightly modified:

A weighted sum of the region-level loss $\mathcal{L}_r$ and the POI-level loss $\mathcal{L}_p$:

$$\mathcal{L} = \alpha \mathcal{L}_r + \beta \mathcal{L}_p$$

where α and β are hyperparameters that control the trade-off between the two losses. After computing the outputs from the Spatial-Temporal Graph Attention Network (ST-GAT), we concatenate the output either with the embedding of the user's centroid (HMT-GRN-C) or/and with the embeddings of the top- k similar users, resulting in a new loss function for the POI-level task:

$$\mathcal{L}_{\text{poi}} = -\sum_{i=1}^{N_{\text{train}}} \log(p_i)$$

where p_i is defined as follows for the two different models:

$$p_i = \text{MLP}(\text{LSTM}_o, \mathbf{c}_u)$$

$$p_i = \text{MLP}(\text{LSTM}_o, \mathbf{c}_u, \text{avg}(\mathbf{c}_{u_j}))$$

where LSTM_o is the output of the LSTM.

Finally, β is set to have a higher value than α because the modifications we have introduced focus on refining the POI-level task. Since we are adding additional information compared to the region-level tasks, we want to increase the weight of the POI-level task accordingly.

5. Dataset

We evaluate our models on two datasets: Gowalla and Foursquare NYC.

The Gowalla dataset [21] is a comprehensive collection of user check-in data captured from the Gowalla location-based social networking service, where users shared their locations by checking into various points of interest (POIs). This dataset is particularly valuable for research in geographic information systems, urban planning, and recommender systems due to its rich set of features which include precise geographic coordinates, timestamps, and user-generated content. Each record in the dataset consists of an user ID, a timestamp, and the geographical coordinates (latitude and longitude) of the check-in location, the venue's name and category (providing deeper insights into user preferences and behaviors).

The Foursquare New York City dataset [22] provides a highly detailed account of user activity within a dense urban environment, encapsulating a wealth of geospatial and

temporal data essential for various advanced analytical applications. It includes long-term (10 months) check-in data in New York City collected from Foursquare from 12 April 2012 to 16 February 2013. This dataset is derived from user interactions with the Foursquare application, where users perform check-ins at diverse points of interest (POIs) throughout New York City, creating a rich tapestry of urban mobility and interaction patterns. Each entry in the dataset typically includes the anonymized user ID, the geographic coordinates (latitude and longitude), the venue ID, a timestamp, venue name and category.

5.1. Dataset preprocessing

For preprocessing, we consider users with visit counts between 20 and 50 in the datasets, then removing POIs visited by less than 10 users. For training and testing, we similarly use the first 80% visits and the last 20% visits of each user’s sequence respectively, after sorting the timestamps in chronological order.

5.2. Graph preprocessing

Our model needs two different types of graphs, the region-level graph and the POI-level graph:

$$G_r = (V_r, E_r)$$

where V_r represents the set of regions and E_r represents the edges between regions. The adjacency matrix A_r is constructed based on spatial proximity, with edges connecting regions that are geographically close.

$$G_p = (V_p, E_p)$$

where V_p represents the set of POIs and E_p represents the edges between POIs within a region. The adjacency matrix A_p captures the transition probabilities between POIs.

6. Experiments

All experiments were performed utilizing the train, validation and test splits obtained as described in 5.1.

We compare different setups utilizing HMT-GRN-C configured with both k-means and Gaussian Mixture Models clustering algorithms to test the effectiveness of the clustering improvement.

Moreover, we test HMT-GRN-C-TK with different combinations and utilize average or concatenation as aggregation techniques and 1 layer or 3 layers of MLP to project embeddings (see Table 1).

For all experiments, we select the 5 most similar users, indeed set top-K equals to top-5. Furthermore we utilized an embedding size of 1024 for user and clustering embeddings and for the aggregate projection embeddings. Furthermore we use the same embedding size for all POI and geoHash embeddings.

All models were trained for 50 epochs with a learning rate of 0.0001 utilizing Adam as an optimizer strategy, and a batch size of 32. All values for dropout were set to 0.8.

As described in 4.5, loss was calculated as the cross-entropy between the predicted Next POI and the true ones.

Finally, we train the original HMT-GRN model on our dataset utilizing the same setting described in its paper,

changing the loss function according to the new one developed by us, in order to compare the performances of our proposed models.

6.1. Metrics

We use standard metric $Acc@K$

$$Acc@K = \frac{1}{N_{test}} \cdot hit@K$$

where $hit@K$ is the number of samples with correct predictions made within the top K of the ranked set for $K \in \{5, 10, 20\}$ and N_{test} is the total number of test samples. It helps to understand the performance of the recommender system for the top K recommendations.

We also use Reciprocal Rank (MRR)

$$MRR = \frac{1}{N_{test}} \cdot \sum_{v=1}^{N_{test}} \frac{1}{rank_v(l_{t_i})}$$

and $rank_v(l_{t_i})$ is the position of the ground truth next POI l_{t_i} in the predicted ranked set for each v -th test sample. MRR gives an overall performance of the ranked set predicted.

7. Results

As a first experiment, we compare the original HMT-GRN model to our HMT-GRN-C models that utilize k-means and Gaussian Mixture Models respectively, in order to observe how the introduction of user cluster centroid would impact the performances of the model. In concordance with our hypothesis, our HMT-GRN-C model shows a slightly increase in all reference evaluation metrics both on Gowalla (+2.7% ACC@5, +3.1% ACC@10, +5.8% ACC@20, +4% MRR) and Foursquare (+2.8% ACC@5, +4.3% ACC@10, +4% ACC@20, +2.4% MRR) datasets. Moreover, we obtain peak performances utilizing our HMT-GRN-C-TK in the configuration that utilizes average as an aggregation strategy with an MLP block for preferences embeddings utilizing 3 layer. We call this configuration Top-G-AVG-3. The Top-G-AVG-3 configuration largely outperforms the HMT-GRN in all reference metrics and further improves the result obtained by the HMT-GRN-C (+7% ACC@5, +7.2% ACC@10, +10.6% ACC@20, +8.5% MRR on Gowalla dataset and +7.4% ACC@5, +9.2% ACC@10, +9.2% ACC@20, +5.9% MRR on Foursquare dataset). These results demonstrate the effectiveness of Clustering when modeling different users preferences with similar behaviours, improving the prediction in Next POI Recommendation. All results are summarized in Table 2, 3.

Additionally, the incremental improvement in accuracy metrics is documented in Figures 2, and 3. These figures demonstrate the consistent enhancements achieved by our models across various testing scenarios, affirming their effectiveness.

Implementation Details

The HMT-GRN model variants leverages the PyTorch deep learning framework for implementation. The region and POI graphs are constructed using the NetworkX library, and the GAT and LSTM are implemented from scratch using the PyTorch library. The model is trained on an NVIDIA GPU to handle the computational complexity of GAT and recurrent updates.

Table 1

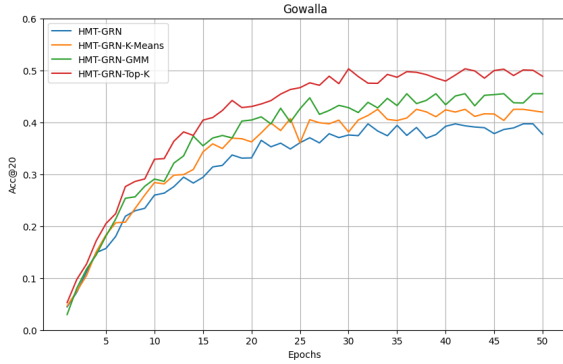
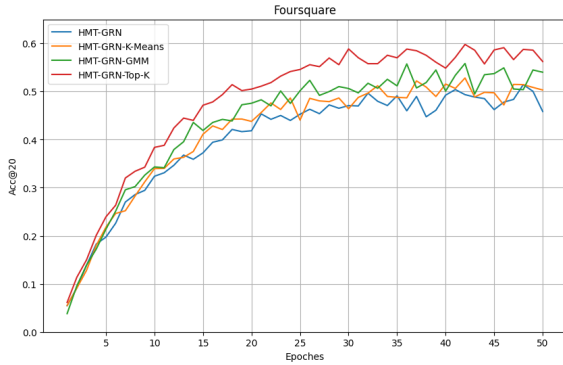
Descriptions for all the experimented setup combinations for the Top-K model.

Top-K Experiments		
Experiment	Aggregation	MLP layers
Top-K-CAT	concatenation	1 - 3
Top-K-AVG	average	1 - 3

Table 2

Results on Gowalla Dataset

Gowalla				
Model	Acc@5	Acc@10	Acc@20	MRR
HMT-GRN	0.275	0.332	0.397	0.203
HMT-GRN-K-Means	0.283	0.342	0.425	0.235
HMT-GRN-GMM	0.302	0.363	0.455	0.243
HMT-GRN-Top-K	0.345	0.404	0.503	0.288

**Figure 2:** Plot of Acc@20 on Gowalla Dataset**Figure 3:** Plot of Acc@20 on Foursquare Dataset**Table 3**

Results on Foursquare Dataset

Foursquare				
Model	Acc@5	Acc@10	Acc@20	MRR
HMT-GRN	0.336	0.410	0.497	0.249
HMT-GRN-K-Means	0.341	0.417	0.512	0.261
HMT-GRN-GMM	0.364	0.453	0.537	0.273
HMT-GRN-Top-K	0.410	0.502	0.589	0.308

8. Conclusion

In this paper, we have presented a novel architecture for next POI recommendation, leveraging two Graph Neural Networks (GNNs) to separately handle spatial and temporal data. Additionally, we incorporated user clustering, which provides an embedding of cluster centroids as an additional feature for the model. This approach captures shared POI preferences among users within the same cluster, improving the accuracy of the next POI prediction. Furthermore we enhanced the improvements obtained incorporating the top-K most similar users embedding, reaching even better performances.

Our experiments on real-world datasets demonstrate that our model outperform the state-of-the-art model proposed in[4], which itself outperforms many other existing models.

The versatility of our model allows it to be applied to other similar tasks, such as route prediction, where predicting the next step in a user's journey can be crucial. Other potential applications include event recommendation, where users' future activities are predicted based on their past behavior and social network analysis, where predicting future interactions can benefit from the same underlying principles.

8.1. Future Works

Our project presents several potential directions for future work:

- 1. Extended Training Time and Enhanced Computational Resources:** Future work could explore the benefits of training the model for extended periods on more powerful machines with larger and more varied datasets. This could potentially improve the model's accuracy and

robustness.

2. Expanding Model Parameters and Testing New Architectures: Another avenue for future research involves increasing the model parameters and experimenting with new emerging architectures for temporal dependencies beyond LSTMs. Examples of such architectures include the Memory-Augmented Neural Networks (MANN) and the Linear-Time Sequence Modeling with Selective State Spaces (MAMBA)[23, 24].

3. Exploring Different Clustering Techniques: While we utilized k-means and Gaussian Mixture Models for user clustering, future studies could apply different clustering techniques, such as or advanced methods like the data filtering approach for cluster analysis using ℓ_0 -norm regularization proposed by Cristofari[25]. This method can be applied in conjunction with the previously mentioned algorithms as it performs data filtering, providing an excellent starting point for clustering. Additionally, it removes the need to manually set the number of clusters k as it automatically determines the optimal value for k .

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